

Scutellaria baicalensis stem and leaf combat chicken-derived respiratory bacterial infection

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ABSTRACT

The Chinese poultry industry has witnessed rapid development, with laying hens playing a pivotal role. However, the escalating demand has led to an exponential increase in the population of laying hens raised, resulting in emerging challenges. Particularly during bacterial infections, substantial losses can be incurred. Presently, most farms heavily rely on antibiotics for disease prevention and control. Although this approach has yielded positive outcomes, prolonged usage may lead to the emergence of drug-resistant strains and residues. Consequently, research on alternative drugs has been initiated due to antibiotic prohibition and growing pathogen resistance. Chinese herbal medicine holds significant prominence across various domains, including animal husbandry and disease treatment, owing to its traditional roots in China. *Scutellaria baicalensis* is a traditional Chinese medicine derived from the dried root of the labiate family plant *Scutellaria baicalensis* that possesses bitter taste and cold properties while exhibiting effects such as heat clearance, dampness elimination, lung purification, fire expulsion and heat detoxification. The aboveground components of *Scutellaria baicalensis* encompass stems and leaves, which yield approximately three times more than their root counterparts as traditional Chinese medicine resources. *Scutellaria baicalensis* products have been successfully applied in animal husbandry with therapeutic effects against sore throat, respiratory diseases, and heat detoxification. Therefore, in pursuit of economic sustainability, this study aims at developing an extract from *Scutellaria baicalensis* stems and leaves for treating respiratory bacterial infections among laying hens. The findings indicate that this extract exhibits excellent therapeutic efficacy against respiratory diseases among laying hens by reducing inflammatory cell levels within their lungs.

1. Introduction

In the current intensive breeding process, the high feeding density, inadequate environmental hygiene, and insufficient air circulation adversely impact the immune system of laying hens, resulting in the disruption of their respiratory mucosal barrier [1]. Consequently, this creates a favorable environment for the proliferation and transmission of various pathogens, ultimately leading to outbreaks of respiratory diseases among laying hens [2]. The primary etiologies of respiratory diseases encompass bacterial and viral infections, as well as secondary infections resulting from factors such as feeding and management

practices [3,4]. Additionally, mixed infections involving diverse bacteria or viruses can induce respiratory symptoms like coughing, nasal discharge, and rales in laying hens, consequently leading to reduced productivity. Acute infectious diseases also pose a significant threat to the survival of affected chickens due to their high infectivity and mortality rates [5–7]. Chronic diseases have a slow but difficult to eradicate condition, persist for a long time, leading to a continuous decline in egg production efficiency and causing serious economic losses [8]. After the outbreak of the disease, the commonly used drugs in clinical practice are mainly antibiotics and chemicals. Even some preventive drugs contain chemical components. If taken for a long time, it will develop a certain

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degree of resistance [9]. Globally, antimicrobial resistance in a variety of clinical bacteria is on the rise, and it is important to explore the use of Chinese medicine in the treatment of clinical microbial infections [10, 11]. Therefore, this study applied the theories of traditional Chinese veterinary medicine. Through the dialectical analysis of traditional Chinese veterinary medicine, targeted treatment of respiratory diseases in laying hens is carried out. Unlike targeted treatment for pathogens, it focuses more on symptoms and is not limited to a single pathogenic pathogen, making it more comprehensive.

Chinese herbal medicine, as a traditional Chinese medicine, has been passed down for thousands of years [12]. Natural Chinese herbal medicine is rich in various chemical components, including not only water, sugars, proteins, fats, and other self-nutrients, but also terpenoids, flavonoids, alkaloids, steroids, wood index, minerals, and other plant secondary metabolites [13]. Flavonoids are common secondary products in plants and are commonly found in various natural plants. In recent years, due to their various effects, such as antibacterial, anti-inflammatory and antioxidant, they have been widely applied in various fields [14]. The *Scutellaria baicalensis* stem and leaf are the aboveground parts of the traditional Chinese medicine *Scutellaria baicalensis*, mainly containing flavonoids and phenolic acid compounds [15]. They have similar active ingredients and are inexpensive compared to roots [16]. At present, *Scutellaria baicalensis* has mature products applied in animal husbandry, which have extensive medicinal value in the aspect of respiratory tract infection, such as treating sore throat, respiratory system disease, clearing away heat and detoxicating [17,18]. The core of our research is to deeply explore the pathogenesis of clinical respiratory pathogens in laying hens, and combine the known pharmacological effects and active components of the *Scutellaria baicalensis* roots to analyze the pharmacological properties of *Scutellaria baicalensis* stems and leaves and the active ingredients they contain. Therefore, this paper aims to clarify the therapeutic efficacy of *Scutellaria baicalensis* stems and leaves for respiratory diseases in laying hens. In order to develop new medicinal parts of traditional Chinese medicine in the future to improve its utilization rate, and open up a new way for the development and utilization of traditional Chinese medicine resources and the prevention and treatment of poultry diseases.

2. Materials and methods

2.1. Source of *Scutellaria baicalensis* and *Scutellaria baicalensis* stem and leaf

Scutellaria baicalensis (Huang Qin, HQ) and *Scutellaria baicalensis* stem and leaf (Huang Qin stem and leaf, HQSL) were collected from Puwa Township, Fangshan District, Beijing. The extraction process of HQSL extract is 50 % ethanol, 1:30 liquid-solid ratio, extracted 3 times, each time for 2 h.

2.2. Pathogen investigation and pathological dissection

Record the pathological symptoms of diseased chickens in the coop. Randomly select 48 laying hens from the coop for dissection and divide them into 6 groups with 8 replicates in each group. Observe the pathological symptoms of diseased laying hens. Sterile isolation of diseased chicken lungs, spleen, kidneys, fallopian tubes, and trachea tissues for pathological observation.

2.3. Pathogen isolation, purification, and gram stain

Apply the separated organs directly onto the pre poured blood plate, invert the 37 °C constant temperature incubator for 24 h, wait for bacteria to grow, scrape a small number of bacteria, inoculate them in the nutrient broth medium, and incubate them in the 37 °C constant temperature shaking incubator at 220 rpm/min for 12 h. After the bacterial liquid becomes turbid, take a small amount of bacterial liquid, and mark

the bacterial liquid on the blood plate. After inverting the 37 °C constant temperature incubator for 12 h, select a single colony, inoculate in a nutrient broth medium and incubate in a constant temperature shaking incubator at 37 °C at 220 rpm/min for 8 h to obtain a purified bacterial solution. Observe the bacteria under a microscope after Gram staining.

2.4. Biochemical experiment of pathogenic bacteria

Using a bacterial micro biochemical identification tube for bacterial biochemical identification, the direct inoculation method is used. Purified and separated bacterial colonies are selected from the cultivated blood plate using an inoculation needle and inoculated into the test tube that needs to be tested. After inoculation, a sealing film is applied and incubated in a 37 °C constant temperature incubator for 24 h.

2.5. Antibiotic sensitivity test for pathogenic bacteria

Adjust the concentration of the cultivated bacterial solution to 1×10^7 CFU mL⁻¹. Take the bacterial liquid and suck 100 µL bacterial solution evenly on a blood agar plate. Attach antibiotic sensitive tablets onto a plate coated with bacterial solution and culture for 12 h. Measure the size of the antibacterial zone.

2.6. The pathogenic bacteria analyzed by PCR

DNA from the pure colony was extracted by bacterial DNA extraction kit (TIANGEN, China) according to the manufacturer's instruction. The DNA were amplified using bacteria universal primers 27F (5'-AGAGTTTGATCCTGGCTCAG-3') and 1492R (5'-ACGGCTACCTGT-TACGACTT-3'). The PCR cycle was as follows: 94 °C for 3 min, followed by 35 cycles at 94 °C for 45 s, 50 °C for 45 s, 72 °C for 90 s, and with a final extension at 72 °C for 7 min. The PCR products were analyzed by electrophoresis, an amplified fragment about 1500 bp was obtained. Send the PCR to Shanghai Biotechnology Co., Ltd. for perform forward and reverse sequencing. Importing the gene sequence of bacteria into the NCBI website in the United States (<http://www.ncbi.nlm.nih.gov>). Using the Blast online software, select the Nucleotide Collection database and run the Blast program to obtain sequences homologous to the gene sequence and corresponding bacterial names. Perform homology analysis using MEGA X.

2.7. In vitro antibacterial test of HQSL

Accurately weigh 1 g of HQSL extract and dissolve it in 10 mL of sterile distilled water. After filtration with a 0.22 µm membrane, take 1 mL of the filtered solution and add it to 9 mL of sterile distilled water for gradient dilution. Repeat the above steps to obtain 100 mg/mL, 10 mg/mL, 1 mg/mL, and 0.1 mg/mL HQSL extracts for use. Adjust the purified pathogen solution to a concentration of 1×10^7 CFU mL⁻¹. Mix different concentrations of HQSL in a 1:1 ratio with pathogenic bacteria. Add an equal volume of culture medium to the control group. Incubate in a constant temperature shaker at 121 rpm/min and 37 °C for 12 h. Take 100 µL of incubated bacterial solution and add it to 900 µL of culture medium for gradient dilution. Repeat the above steps and dilute the co incubated bacterial solution by 10^1 - 10^8 times. Take 10 µL of diluted bacterial solution and drop them onto blood plates. After drying, incubate at 37 °C.

2.8. Animals and treatments

This study was carried out in Beijing Shuangyin Farm (Pinggu, Beijing, China). One hundred and twenty-eight HY-LINE VARIETY BROWN laying hens were obtained at the 360-day-old in Beijing Shuangyin Farm. These hens suffer from respiratory symptoms such as coughing, runny noses, etc. These laying hens were randomly divided into four groups, including the control group, HQSL powder, HQ powder and

HQSL extract groups with 4 replicates in each group and 8 laying hens in each replicate. The control group was only fed with basic diet, at the same time, the HQSL powder group was fed with diet supplemented with 760 mg/kg (kg of the feed) HQSL powder, the HQ powder group was fed with diet supplemented with 190 mg/kg (kg of the feed) HQ powder and the HQSL extract group was fed with diet supplemented with 125.56 mg/kg (kg of the feed) HQSL extract. The experimental period was 15 days, with pre feeding for 3 days and formal feeding for 12 days. The experimental laying hens were raised in three-tier cages. During the trial, all laying hens were fed four times daily and had free access to diet and water. Collect eggs at 11:30 every day for inspection and recording. Observe and monitor the status of laying hens daily. (Fig. 3A).

2.9. Sample collection

Collect eggs every day after the start of the experiment and record the number of eggs produced and abnormal eggs in each group. Observe and monitor the health status, feed intake, and fecal excretion of laying hens, and observe the clinical symptoms and pathological changes of laying hens. On day 0 and day 12 of the experiment, 8 laying hens were randomly selected from each group. After serum sampling from wing vein, the laying hens were euthanized by pentobarbital sodium. Anatomy under sterile conditions, collect lungs, trachea, fallopian tubes, liver, and spleen.

2.10. Pathological sections of lung tissue

Fix the dissected lung tissue in 4 % paraformaldehyde solution for 24 h. Remove the fixed tissue from 4 % paraformaldehyde. Take the obvious lesion area, cut it into moderately sized tissue blocks and place them in an embedding box. Rinse overnight with tap water in a beaker. After overnight washing, the tissue was subjected to ethanol gradient dehydration, soaking in 70 % ethanol solution for 1 h, 80 % ethanol solution for 40 min, 90 % ethanol solution for 40 min, 95 % ethanol solution for 30 min, 100 % anhydrous ethanol solution for 30 min, and 100 % anhydrous ethanol solution for another 30 min to completely dehydrate the tissue. Soak the dehydrated tissue in a mixture of xylene and anhydrous ethanol at a ratio of 1:1 for 20 min. Repeat twice for 20 min in 100 % xylene solution for tissue transparency treatment. Put the transparent treated tissue into an embedding machine for tissue embedding and store the embedded tissue at 4 °C. Cut the tissue blocks into thin slices using a slicer and dry them overnight in a 37 °C oven using a spreading machine.

The dried tissue slices are subjected to dewaxing treatment. Repeat twice for 10 min in xylene solution, with xylene: anhydrous ethanol = 1:1 mixed solution for 5 min. Repeat twice for 5 min in anhydrous ethanol solution, with 95 % ethanol solution for 5 min, 80 % ethanol solution for 5 min, and 70 % ethanol solution for 5 min. After 5 min in distilled water, remove and perform hematoxylin eosin staining. Soak in hematoxylin staining solution for 10 min and rinse with running water for 5 min to remove the blue color from the slices, After 1 s of color separation with 1 % hydrochloric acid alcohol, rinse with tap water for 40 min, soak in 70 % ethanol solution for 5 min, soak in 80 % ethanol solution for 5 min, stain with eosin for 1 min, rinse with tap water until all staining is removed, soak in 95 % ethanol solution for 1 min, soak in anhydrous ethanol for 1 min, repeat twice. Transparent the stained sections and place them in a mixture of anhydrous ethanol and xylene at a ratio of 1:1 for 2 min. Soak the xylene solution for 5 min, repeat twice, and then air dry naturally. Seal the sections with neutral gum and air dry them at room temperature for 24 h. Then observe the samples under an optical microscope.

2.11. Laying performance and egg quality

The production of laying hens in each group was recorded. In addition, the egg production rate and abnormal egg rate were calculated

throughout the experiment, according to the following formula:

$$\text{Egg production rate} = \frac{\text{Total number of eggs}}{\text{Total number of laying hens}} \times 100\%$$

$$\text{Abnormal egg rate} = \frac{\text{Total number of abnormal eggs}}{\text{Total number of eggs}} \times 100\%$$

2.12. Analysis of HQSL extract by ultra-high performance liquid chromatography tandem mass spectrometry (UHPLC-MS/MS)

Sample processing: Weigh 50 mg of sample and add 50 μ L 75 % methanol aqueous solution, vortex for 60 s, sonicate for 15 min, centrifuge 17000 g for 15 min, take the supernatant, and vacuum dry. Join 100 μ L 50 % methanol aqueous solution, vortex for 60 s to fully dissolve. Centrifuge 17000 g for 15 min, take the supernatant and pass it through a 0.22 μ m microporous filter membrane. Place it in an injection bottle and perform UHPLC-MS/MS analysis. Blank samples were treated under the same conditions.

Liquid phase conditions: Chromatographic column: ACQUITY UPLC HSS T3 1.8 μ m 2.1 $\times$ 100 mm; column temperature: 35 °C; injection volume: 5 μ L; velocity of flow: 400 μ L/min; positive ion mode mobile phase: A (deionized water, containing 0.1 % formic acid) B (acetonitrile, containing 0.1 % formic acid); negative ion mode mobile phase: A (Water, containing 2 mM ammonium acetate) B (acetonitrile). The elution procedure of UHPLC-MS/MS mobile phase is shown in Table 1.

Mass spectrometry conditions: Perform mass spectrometry analysis using a mass spectrometer (Agilent 6545A QTOF) and control software (LC/MS Data Acquisition, Version B.08.00). Select fully automatic MS/MS mode for primary and secondary mass spectrometry data collection. Collection rate: MS1 (6 spectrum/s), MS2 (12 spectrum/s) Secondary collision energy samples: 10 V, 40 V. Select 12 ions from the primary spectrum for secondary scanning. Quality scanning range: first level m/z (50–1600), second level m/z (20–1600).

The ESI ion source parameters are set as follows: ion source temperature: 320 °C, nitrogen flow rate: 8 l/min, sheath gas flow rate: 12 l/min, sheath gas temperature: 350 °C; Capillary voltage: 4000 V (positive ion mode), 3500 V (negative ion mode).

2.13. Statistical analysis

All data were shown as means $\pm$ standard deviation and analyzed through using GraphPad Prism 9 (GraphPad Software, USA). Significant differences among groups were conducted using *t*-test by one way analysis of variance (ANOVA). The results were assumed to be significant when $P < 0.05$. MSDIAL4.60 software for chemical composition and activity analysis of traditional Chinese medicine.

3. Results

3.1. Clinical manifestations of respiratory diseases in laying hens

We randomly observed 3 groups of 60 diseased laying hens in the chicken coop of the farm, according to clinical symptom observation, the main clinical symptoms of these laying hens include coughing, runny nose, and breathing with a "snoring" sound. Abnormal egg production in sick chickens, including spotted eggs, soft shell eggs, and small eggs (Fig. 1A–B). The pathological examination results indicate that some laying hens have ruptured yolks and yolk peritonitis. There is yellow secretion in the trachea and multiple areas of bleeding. Cloaca edema accompanied by prolapse. (Fig. 1C–F).

3.2. Clinical pathogen isolation and identification of respiratory diseases in laying hens

Pathogenic bacteria do not grow on regular agar, but on fresh blood

Table 1
Scutellaria baicalensis stem and leaf flavonoids.

Name	Molecular formula	Molecular weight	Peak time	Peak area/min	Additive ion	Relative content
Scutellarin	C ₂₁ H ₁₈ O ₁₂	462.366	9.143	22521065.35	[M+H] ⁺	10.15 %
Delphinidin 3-O-glucoside	C ₂₁ H ₂₀ O ₁₂	465.1028	9.17725	13092448	[M+H] ⁺	5.898 %
Apigenin	C ₁₅ H ₁₀ O ₅	269.0455	14.61887	8418032	[M - H] ⁻	3.792 %
Apigenin 7-glucoside	C ₂₁ H ₂₀ O ₁₀	863.204	14.02768	6596218.5	[2M - H] ⁻	2.972 %
Luteolin	C ₁₅ H ₁₀ O ₆	287.0555	12.48397	6349336	[M+H] ⁺	2.860 %
Flavanonol	C ₁₅ H ₁₂ O ₇	305.0656	7.875216	5816410	[M+H] ⁺	2.620 %
Chrysin	C ₁₅ H ₁₀ O ₄	253.0506	17.5064	3207913.25	[M - H] ⁻	1.445 %
Kaempferol 3-O-sophoroside	C ₂₇ H ₃₀ O ₁₆	609.1461	9.1044	3138662.75	[M - H] ⁻	1.414 %
Plantagoside	C ₂₁ H ₂₀ O ₁₁	449.1079	8.749884	3014625.25	[M+H] ⁺	1.358 %
Acacetin	C ₁₆ H ₁₂ O ₅	285.0758	17.50537	2867765.5	[M+H] ⁺	1.292 %
Esculetin	C ₉ H ₆ O ₄	177.0185	6.707883	2826688.5	[M - H] ⁻	1.273 %
Aloin-A	C ₂₁ H ₂₂ O ₉	419.1337	19.08088	2858708	[M+H] ⁺	1.288 %
Isoschaftoside	C ₂₆ H ₂₈ O ₁₄	565.1552	6.845284	2679468.25	[M+H] ⁺	1.207 %
4',5,7-Trihydroxyflavone	C ₁₅ H ₁₀ O ₅	269.0458	15.13595	2671360.5	[M - H] ⁻	1.203 %
Pinocembrin	C ₁₅ H ₁₂ O ₄	257.0809	17.44078	2573273.25	[M+H] ⁺	1.159 %
Isoquercitrin	C ₂₁ H ₂₀ O ₁₂	927.1837	9.25775	2547067.75	[2M - H] ⁻	1.147 %
Hyperoside	C ₂₁ H ₂₀ O ₁₂	927.1837	10.34255	2386881	[2M - H] ⁻	1.075 %
Acacetin 7-Rutinoside	C ₂₈ H ₃₂ O ₁₄	593.1865	11.88003	2321230.5	[M+H] ⁺	1.046 %
Baicalin	C ₁₅ H ₁₀ O ₅	271.0601	14.5932	2126961.5	[M+H] ⁺	0.958 %
Quercetin	C ₁₅ H ₁₀ O ₇	301.0354	10.34255	1169607.875	[M - H] ⁻	0.527 %
Kaempferol	C ₁₅ H ₁₀ O ₆	287.055	9.58015	876813.9375	[M+H] ⁺	0.395 %
Quercetin 3- methyl ether	C ₁₆ H ₁₂ O ₇	317.0656	11.19185	695645.8125	[M+H] ⁺	0.313 %
Vitexin-2-O-rhamnoside	C ₂₇ H ₃₀ O ₁₄	579.1708	8.717216	621310.5	[M+H] ⁺	0.280 %
5,8-dihydroxyflavone	C ₁₅ H ₁₂ O ₄	255.0636	17.67308	607745.5	[M - H] ⁻	0.274 %
Baicalin	C ₂₁ H ₁₈ O ₁₁	891.1625	12.55202	422410.4375	[2M - H] ⁻	0.190 %
Daidzin	C ₂₁ H ₂₀ O ₉	417.118	12.23422	413858.0313	[M+H] ⁺	0.186 %
Oroxylin A	C ₁₆ H ₁₂ O ₅	283.0612	17.5092	386420.4375	[M - H] ⁻	0.174 %
Eriodictyol	C ₁₅ H ₁₂ O ₆	575.1195	12.70092	365481.3438	[2M - H] ⁻	0.165 %
Hesperetin	C ₁₆ H ₁₄ O ₆	301.0718	14.48653	268771.5625	[M - H] ⁻	0.121 %
Myricetin	C ₁₅ H ₁₀ O ₈	317.0303	10.34255	201429.25	[M - H] ⁻	0.091 %
Liquiritin	C ₂₁ H ₂₂ O ₉	417.1191	12.23983	194515.4219	[M - H] ⁻	0.088 %
Rhoifolin	C ₂₇ H ₃₀ O ₁₄	577.1563	9.03225	156132.3594	[M - H] ⁻	0.070 %
Neohesperidin dihydrochalcone	C ₂₈ H ₃₆ O ₁₅	611.1981	9.364984	86959.85938	[M - H] ⁻	0.039 %

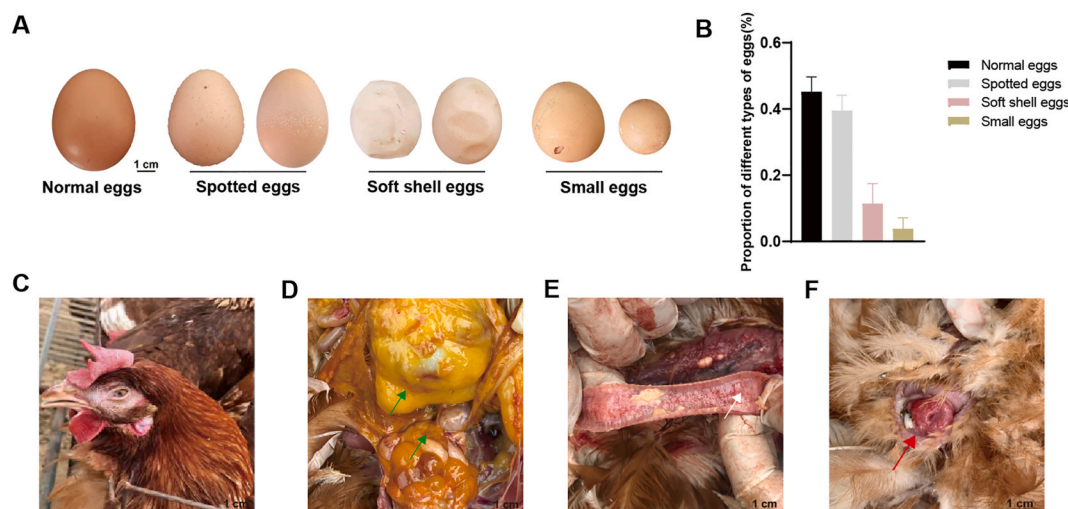


Fig. 1. Clinical and pathological manifestations of diseased chickens. (A) Compared to normal eggs, diseased hens produce spotted eggs, soft shell eggs, and small eggs. Scale bar = 1 cm. (B) The proportion of 60 sick laying hens laying normal eggs, spotted eggs, soft shell eggs and small eggs. (C) Sick hens are listless and lethargic. (D) Pathological changes in the abdominal cavity. Oviduct edema, thickening, peritoneal inflammation, yolk rupture and yolk peritonitis can be seen in the reproductive tract (Green Arrow). Scale bar = 1 cm. (E) Pathological changes of the trachea. Hemorrhagic spots visible in the trachea (White Arrow). Scale bar: 1 cm. (F) Pathological changes in the cloaca. Swelling and prolapse of the cloaca (Red Arrow). Scale bar = 1 cm. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

agar medium, they grow small colonies with a smooth surface and gray color. The colonies are surrounded by a micro blood ring, and the bacterial body is gray white and transparent (Fig. 2A). After adding 5 % fetal bovine serum to regular agar, pathogenic bacteria can grow small colonies with a yellow surface (Fig. 2B). Through Gram staining observation, it was found that the bacteria were highly stained red at both ends, indicating that they were Gram negative bacillus (Fig. 2C). The

extracted pathogen DNA was identified by 16S rRNA. After PCR amplification and 1 % agarose gel electrophoresis, there were obvious bands at 1500 bp, which was consistent with the expected results. The sequenced sequence was subjected to blast analysis in GenBank of NCBI, and the results showed that the pathogenic bacteria were highly like *Gallibacterium anatis* with a nucleotide homology of up to 99.84 %. The gene sequences with the top 50 similarities were imported into MEGA-X

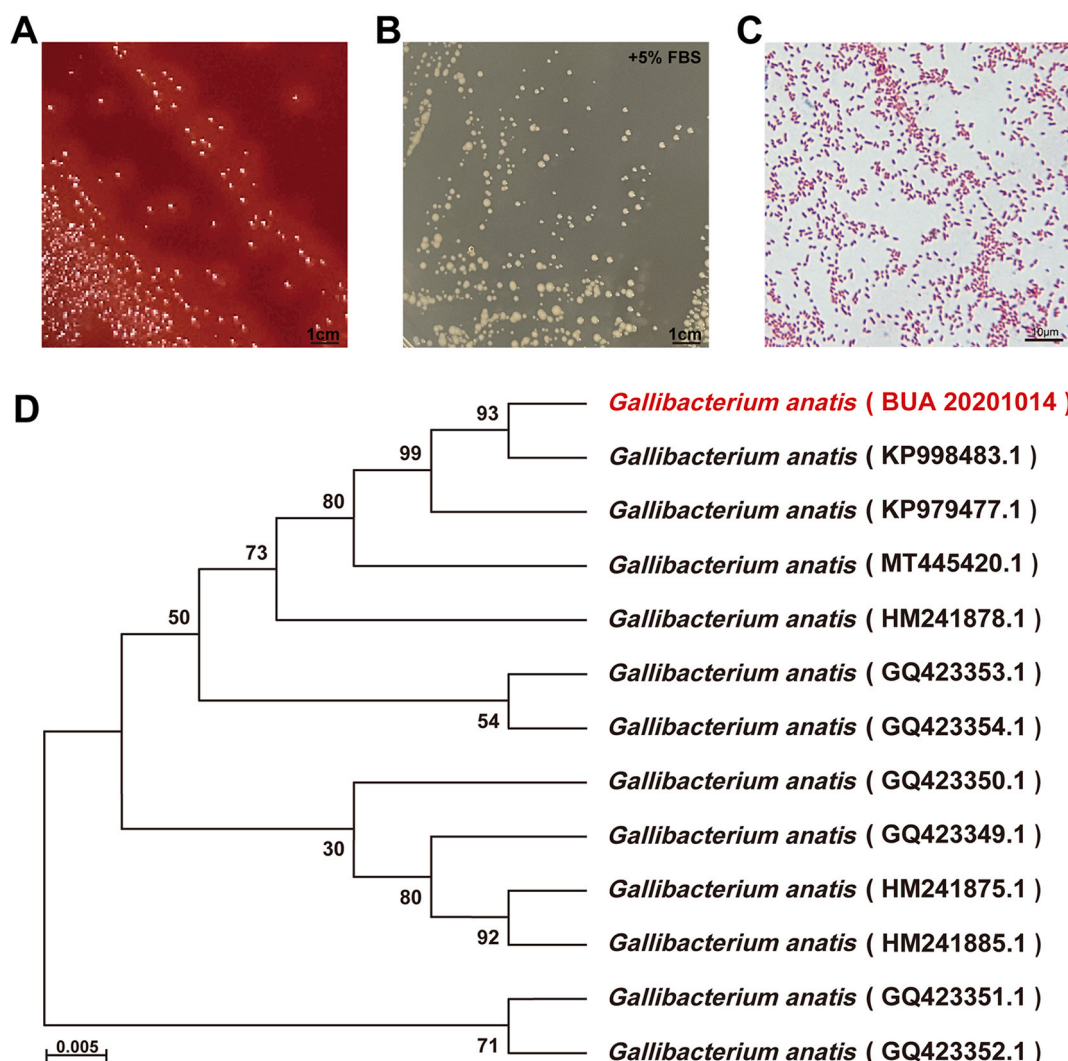


Fig. 2. Isolation and identification of *Gallibacterium anatis*. (A) Pathogenic bacteria grow on blood agar medium, with a smooth and gray, white transparent surface. Scale bar = 1 cm. (B) Pathogenic bacteria cannot grow on ordinary LB agar, but they can grow gray colonies on LB agar with 5 % FBS added. Scale bar = 1 cm. (C) Gram-stained morphology of bacteria. Observation shows that Gram negative isolated bacteria are strongly stained with red at both ends. Scale bar = 10 μm. (D) After genetic homology analysis, it was found that the isolated pathogenic bacteria had a 93 % homology with reference strain (KP998483.1), belonging to the same genetic evolutionary branch. It was determined that the pathogenic bacteria were *Gallibacterium anatis*. Note: Numbers in brackets are the accession numbers of sequences in GenBank. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

for gene homology analysis. After comparison, the isolated *Gallibacterium anatis* had a 93 % homology with the reference strain (KP998483.1) and were in the same genetic evolutionary branch. It was determined that the pathogenic bacterium was *Gallibacterium anatis* (Fig. 2D).

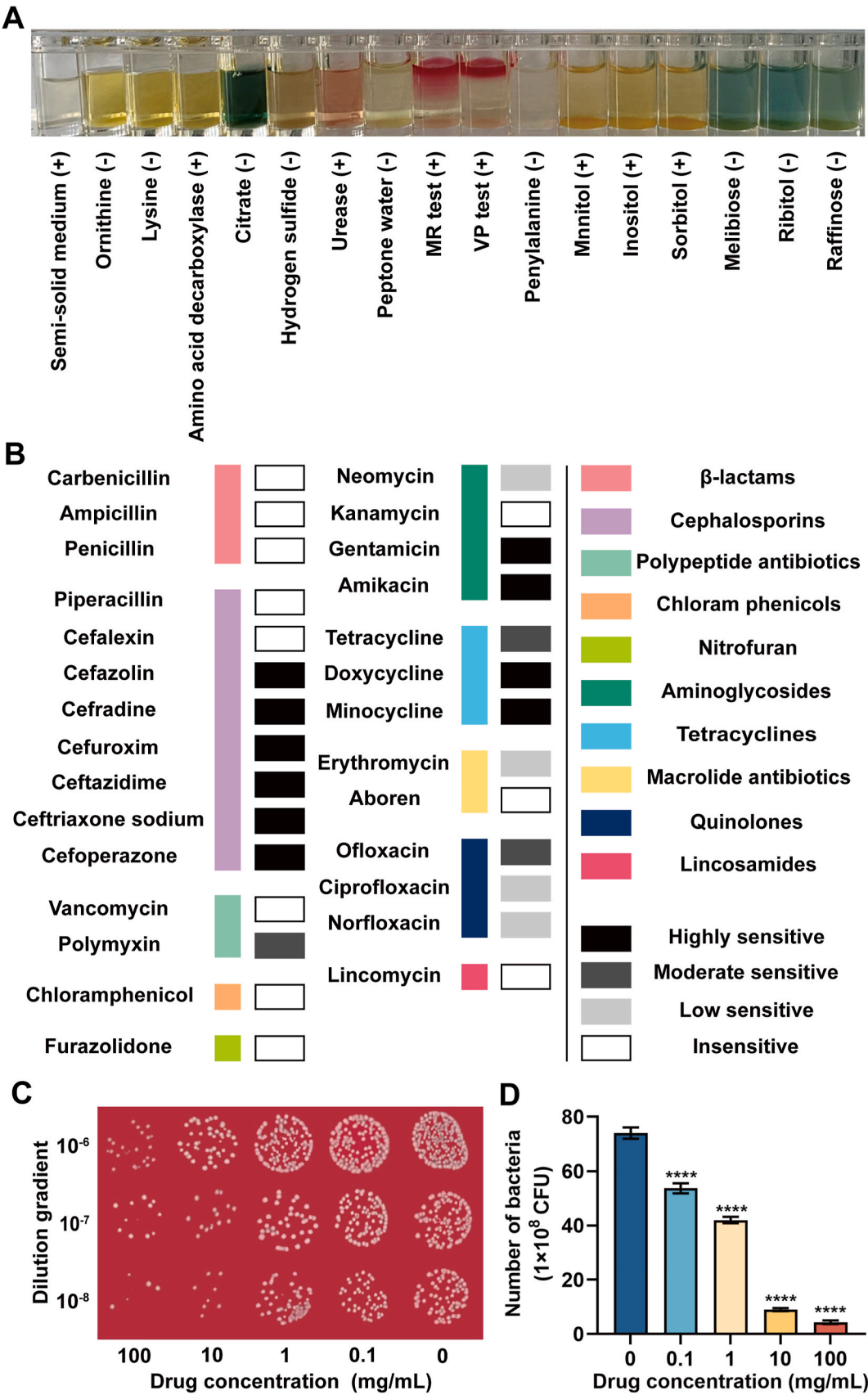
3.3. The characteristics of *Gallibacterium anatis* and the antibacterial activity of *Scutellaria baicalensis* stem and leaf extracts

Through biochemical identification of *Gallibacterium anatis* found that it can ferment glucose, mannitol, inositol, and sorbitol. The peroxidase and oxidase tests were positive (Fig. 3A). The results of the drug sensitivity test showed that *Escherichia coli* was not sensitive to ampicillin, carbenicillin, penicillin, piperacillin, cephalosporin, vancomycin, compound sulfamethoxazole, chloramphenicol, kanamycin, midecamycin, lincomycin, and furazolidone; moderately sensitive to polymyxin, tetracycline, and ofloxacin; low sensitivity to neomycin, erythromycin, ciprofloxacin, and norfloxacin; highly sensitive to cefazolin, cefradine, cefuroxime, ceftazidime, ceftriaxone, cefoperazone, gentamicin, amikacin, doxycycline, and minocycline. Cephalosporin and tetracycline antibiotics have a good inhibitory effect on this

pathogenic bacterium, which has a significant impact as β-Lactamide antibiotics have strong resistance (Fig. 3B). To investigate the antibacterial ability of HQSL extract against *Gallibacterium anatis* in vitro, the *Gallibacterium anatis* were cocultured with HQSL extract. The results showed that the extract of HQSL had extract an inhibitory effect on *Gallibacterium anatis* at a concentration of 100 mg/mL to 1 mg/mL, with an inhibition rate of over 50 %. The concentration of 0.1 mg/mL had an inhibitory effect on *Gallibacterium anatis*, but the inhibitory effect was higher than 50 %, indicating poor antibacterial effect (Fig. 3C and D).

3.4. The therapeutic effect of *Scutellaria baicalensis* stem and leaf on respiratory diseases in laying hens

Observe the clinical symptoms of laying hens after drug treatment. Respiratory symptoms have shown significant effects, with runny nose and head shaking symptoms basically cured. The therapeutic effect of HQSL extract is the best among them. The egg production rate of the control group showed a decreasing trend within 0–12 days. Except for the control group, the egg production rate of all medication groups showed an upward trend. The group of HQSL extracts showed the



(caption on next page)

Fig. 3. The characteristics of *Gallibacterium anatis* and the antibacterial activity of *Scutellaria baicalensis* stem and leaf extracts. (A) Bacterial biochemical identification results. *Gallibacterium anatis* can ferment glucose, mannitol, inositol, and sorbitol. The catalase and oxidase tests were positive. **(B)** Antibiotic susceptibility test of *Gallibacterium anatis*. Cephalosporins and tetracyclines have good inhibitory effects on *Gallibacterium anatis*. β -lactam antibiotics have no inhibitory effect on *Gallibacterium anatis*. (20 mm $\geq$ Highly sensitive $\geq$ 15 mm); (14 mm $\geq$ Moderate sensitive $\geq$ 10 mm); (9 mm $\geq$ Low sensitive $\geq$ 1 mm); (Insensitive = 0 mm) **(C, D)** Antibacterial test of *Gallibacterium anatis* with *Scutellaria baicalensis* stem and leaf. *Scutellaria baicalensis* stem and leaf extract has an inhibitory effect on *Gallibacterium anatis* at a concentration of 100 mg/mL to 1 mg/mL, with an antibacterial rate of over 50 %. Concentrations below 1 mg/mL have antibacterial effects on *Gallibacterium anatis*, but the antibacterial effect is less than 50 %, indicating poor antibacterial effect (**** $P < 0.0001$). All experiments were repeated three times and yielded consistent results.

highest growth (Fig. 4B). The abnormal egg rate in each group showed an increasing trend in the first 3 days of the experiment. The abnormal egg rate in the control group continued to increase after 3 days, while the medication group slowly decreased after 3 days of the experiment (Fig. 4C). After 12 days of treatment, there was a significant increase in egg production rates between each medication group and the control group, while the difference in abnormal egg rates was significantly reduced. Among them, the HQSL extract group has the most significant effect (Fig. 4D and E). Observation of changes in the lungs of laying hens in different treatment groups. Before treatment, there was significant congestion and parenchymal lesions in the lung tissue of laying hens. Compared to before treatment, the untreated group had more significant lung lesions. After treatment with HQSL powder, HQ powder, and HQSL extract, there was a significant improvement in lung tissue lesions. Among them, the HQSL extract has the best therapeutic effect after treatment (Fig. 5A and B). Observation of changes in the lungs of laying hens in different treatment groups through H&E staining. Before treatment, there was significant infiltration of inflammatory cells in the lung tissue of laying hens observed under a microscope, accompanied by inflammatory reactions in the lungs. After treatment, the area of inflammatory cells in the control group increased and the lesions became more severe. The medication group showed an increase in alveolar cavity area, a more complete alveolar structure, reduced congestion, and a decrease in inflammatory cells. Compared with before treatment, treatment with HQSL powder, HQ powder, and HQSL extract can significantly reduce the area of inflammatory cells (Fig. 5C and D). Prove that after treatment with three drugs, there is a significant improvement effect on laying hens infected with respiratory diseases.

3.5. Analysis of components in *Scutellaria baicalensis* stem and leaf

By comparing several extraction methods of HQSL water extraction, ultrasonic water extraction, ultrasonic alcohol extraction, aspergillus niger fermentation extraction, and cellulose fermentation extraction, the total flavonoid content was detected [19–23]. It was found that the ultrasonic alcohol extraction method of HQSL had the highest total flavonoid content. On this basis, single factor experiments and orthogonal experiments were conducted to analyze the optimal process for ultrasonic alcohol extraction of HQSL. The optimal process was obtained as follows: 50 % ethanol, a 1:30 liquid-to-material ratio, and extraction 3 times, each time for 2 h. This method is stable and feasible, with a total flavonoid content of 36.92 mg/g and an extraction rate of 18.73 % (Fig. 6A). Through UHPLC-MS/MS and MSDIAL4.60 comparative analysis, a total of 86 chemical components were identified. This includes flavonoids, phenolic acids and organic acids, alkaloids, amino acids, and other components (Fig. 6B–Tables 1–5). The relative content of scutellarin in the top 20 active ingredients of HQSL extract is 10.15 %, which has antitumor and anti-inflammatory effects [24]. The relative content of delphinidin-3-O-glucoside is 5.898 %, which has antioxidant and anti-cancer effects [25]. The relative content of apigenin is 3.792 %, which has antitumor, antiviral, and antibacterial effects [26]. Apigenin-7-O-glycoside, with a relative content of 2.972 %, has the functions of maintaining capillary permeability, antibacterial and anti-inflammatory [27,28]. The relative content of phenolic and organic acid compounds is 23 %, including malic acid, cis aconitic acid, salicylic acid and gallic acid. In addition, it also contains various compounds such as alkaloids, amino acids, lipids, polysaccharides, terpenes, etc. (Fig. 6C

and D).

4. Discussion

Scutellaria baicalensis is a traditional Chinese medicine used to treat lung diseases. It belongs to the category of heat-clearing and dampness-reducing drugs and belongs to the lung meridian. It has significant anti-inflammatory and antiviral effects and has therapeutic effects on pneumonia, lung heat cough, and upper respiratory tract infections [17, 29]. Additionally, *Scutellaria baicalensis* has excellent antibacterial effects and is not easily resistant to drugs [30]. Research has shown that *Scutellaria baicalensis* has the effect of resisting many kinds of bacteria, and the antibacterial effect is better than *Coptis chinensis* [31]. The laying hens in this chicken farm are mainly suffering from respiratory symptoms, including coughing, snoring, runny nose, head shaking, and severe decline in productivity. According to the analysis of traditional Chinese veterinary medicine, it belongs to the damp heat type of disease and is suitable for the symptomatic treatment of *Scutellaria baicalensis*. Therefore, based on the clinical and pathological symptoms of this disease, the *Scutellaria baicalensis*, *Scutellaria baicalensis* stem and leaf and *Scutellaria baicalensis* stem and leaf extracts are used for treatment.

In our study, the *Gallibacterium anatis* was isolated in all ill chickens. Although they might be found occasionally as a part of the normal microbiota of the upper respiratory and lower reproductive tracts of chickens, experimental evidence has shown that under certain conditions associated with debilitation, nutrition and climatic factors, these organisms may singly or in concert with other organisms flare up to cause severe infections with high morbidity and mortality [32,33]. Although some inactivated vaccines against *Gallibacterium anatis* have been developed with a certain degree of success [34], there is still no available vaccine on the market able to confer protection against the most prevalent *Gallibacterium anatis* biovars and strains.

According to the results of in vivo experiments, it can be concluded that in the early stages of infection, *Gallibacterium anatis* multiplies extensively in the trachea of laying hens, causing respiratory symptoms, and leading to a continuous decrease in egg production rates and an increase in abnormal egg rates, which persist for a long time. *Scutellaria baicalensis* stem and leaf have a good effect on the main respiratory clinical symptoms caused by the disease in laying hens in this chicken farm, mainly improving cough and snoring symptoms, slightly improving runny nose and head shaking symptoms, and improving the production performance decline caused by diseases, restoring the production performance of laying hens. Through further analysis of lung pathological sections, it was found that *Scutellaria baicalensis* stem and leaf can also reduce the area of inflammatory cells in the lungs, reduce pulmonary inflammatory reactions, and improve pathological symptoms. When the total flavonoid content is the same, *Scutellaria baicalensis* stem and leaf can improve respiratory symptoms, production performance, and reduce lung inflammatory cell levels. They can also have the same effect as *Scutellaria baicalensis*. However, due to the lower total flavonoid content in *Scutellaria baicalensis* stem and leaf compared to *Scutellaria baicalensis*, the dosage of *Scutellaria baicalensis* stem and leaf is increased.

As an economic animal, the most important thing for laying hens is their production performance and egg quality [35]. After treatment with *Scutellaria baicalensis* stem and leaf, the disease of diseased chickens has been significantly improved, and the production performance of laying

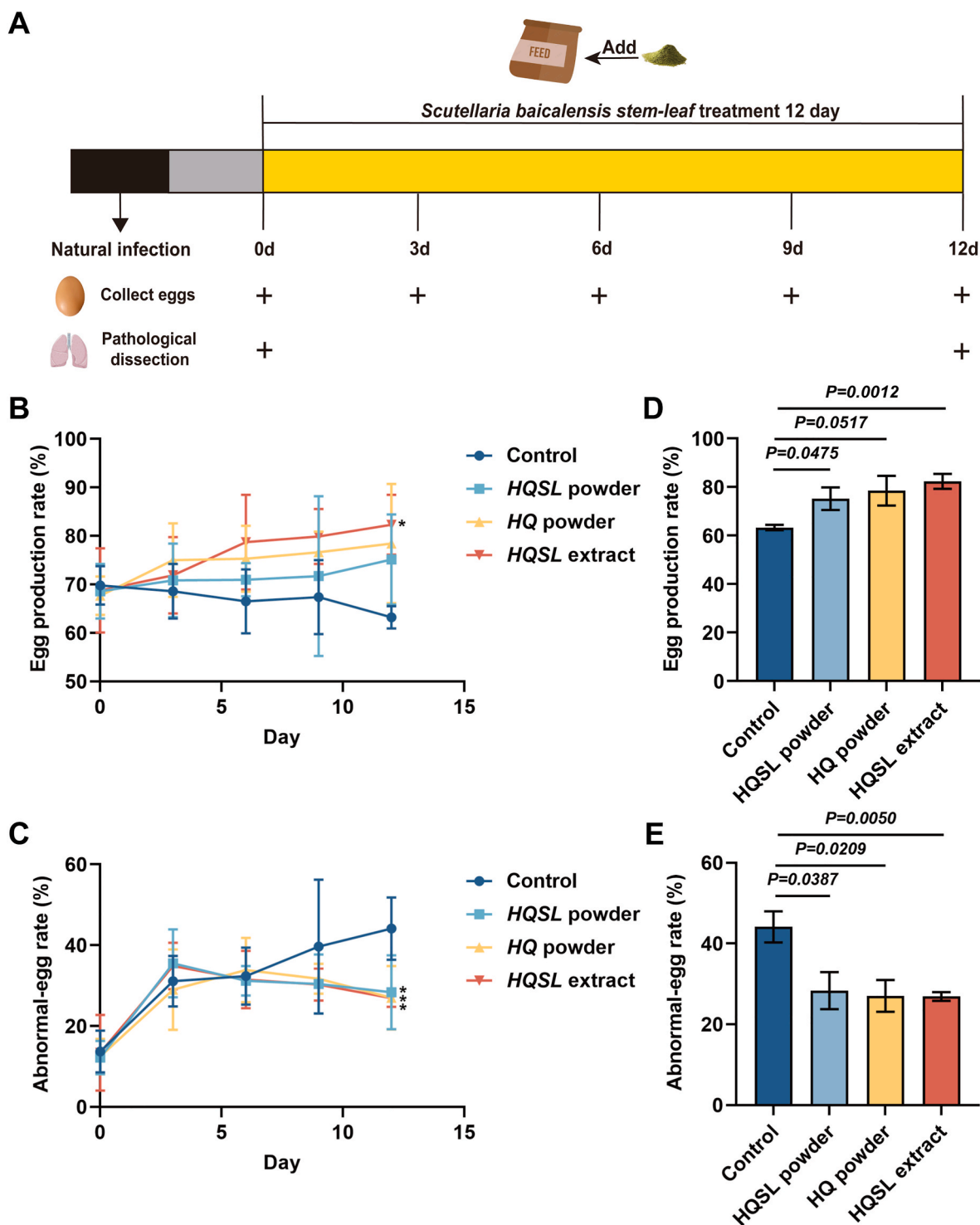


Fig. 4. The therapeutic effect of *Scutellaria baicalensis* stem and leaf extract on *Gallibacterium anatis* infection in laying hens. (A) The experimental scheme of *Scutellaria baicalensis* stem and leaf for the treatment of *Gallibacterium anatis* infection in laying hens. (B) Change of egg production rate within 12 days of experiment. After treatment with *Scutellaria baicalensis* stem and leaf extract, the egg production rate was significantly improved (* $P < 0.05$). (C) Change of abnormal egg rate within 12 days of experiment. After treatment, the abnormal egg rate in all groups significantly decreased (* $P < 0.05$). (D) The egg production rate of each group after 12 days of medication. After treatment, the egg production rate in all groups significantly increased, with the best effect observed in the *Scutellaria baicalensis* stem and leaf extract group. (E) The abnormal egg rate of each group after 12 days of medication. After treatment, the abnormal egg rate in all groups significantly decreased, with the best effect observed in the *Scutellaria baicalensis* stem and leaf extract group.

hens, including egg production rate and abnormal egg production rate, has been improved. The quality of eggs has also improved. Therefore, in this study, the recovery of laying hens from diseases and the improvement of production performance were attributed to the application of

Scutellaria baicalensis stem and leaf, achieving full utilization of resources and reducing production costs. The above results indicate that *Scutellaria baicalensis* stem and leaf have the potential to serve as an excellent drug for treating respiratory diseases in economic animals,

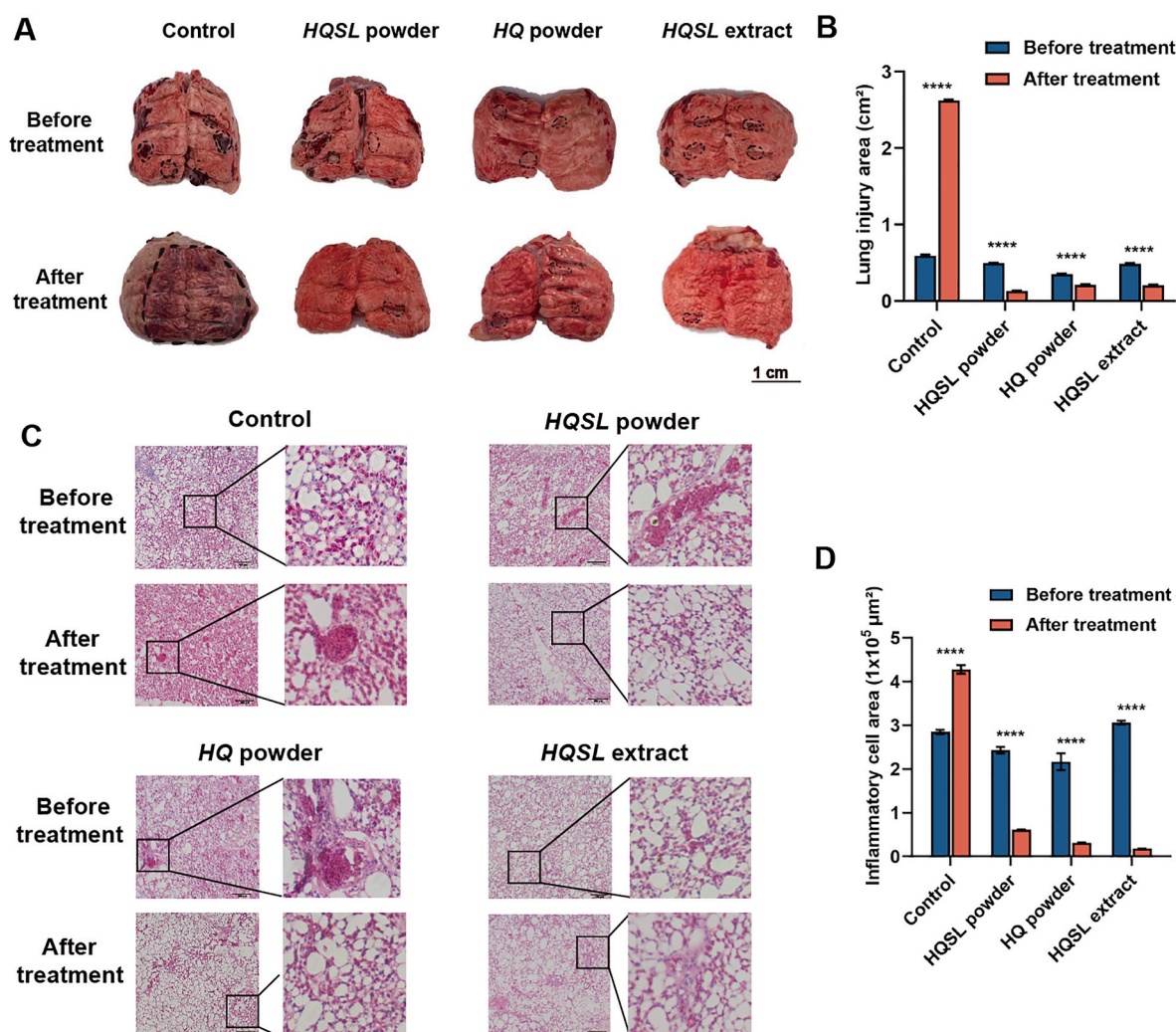


Fig. 5. Recovery effect of *Scutellaria baicalensis* stem and leaf on lung tissue injury. (A) Pathological change of lung tissue before and after treatment in each group. Scale bar = 1 cm. (B) The area of injury in lung tissue. After treatment, all treatment groups were able to significantly reduce the area of lung injury (**** $P < 0.0001$). (C) H&E staining of lung tissue before and after treatment in each group. Scale bar = 100 μm . (D) The area of inflammatory cells in lung tissue. After treatment, all treatment groups were able to significantly reduce the area of inflammatory cells in lung tissue, with the most significant effect observed in the *Scutellaria baicalensis* stem and leaf extract group (**** $P < 0.0001$).

improving production performance, and enhancing economic benefits. The study of *Scutellaria baicalensis* stem and leaf provides an alternative strategy for treating respiratory diseases in laying hens, improving egg quality, and developing new traditional Chinese veterinary medicines.

5. Conclusion

The reason for the illness of the laying hens in the chicken farm this time is due to *Gallibacterium anatis* infection. The *Scutellaria baicalensis* stem and leaf extract has a good antibacterial effect on *Gallibacterium anatis* in the concentration range of 1–100 mg/mL *in vitro*. Adding 125.56 mg/kg *Scutellaria baicalensis* stem and leaf extract to the diet can improve the clinical symptoms of laying hens infected with respiratory diseases, increase egg production rate, reduce abnormal egg rate, alleviate tissue damage caused by *Gallibacterium anatis*, and have the best therapeutic effect on laying hens infected with respiratory diseases. The *Scutellaria baicalensis* stem and leaf contain 86 chemical components, including flavonoids, phenolic acids, organic acids, alkaloids, amino acids, etc. The content of flavonoids is the highest, with a relative content of about 52 %. Among them, baicalin, delphinidin-3-O-glucoside, and apigenin are the main flavonoids with antibacterial and anti-inflammatory effects. These results indicate that *Scutellaria baicalensis*

stem and leaf have a good therapeutic effect on respiratory diseases in laying hens.

Research indicates that the primary constituents of *Scutellaria baicalensis* extract, such as baicalin, may exert their pharmacological effects through various mechanisms. First, *Scutellaria baicalensis* extract may exert immunomodulatory effects by regulating the expression of inflammation-related cytokines. For example, in one study, a decoction of *Scutellaria baicalensis* was found to significantly reduce the levels of interleukin-6 (IL-6) and tumor necrosis factor- α (TNF- α) in the serum of infected mice, suggesting that it may reduce the inflammatory response by inhibiting the inflammatory signaling pathway [36]. In addition, *Scutellaria baicalensis* extract may also enhance the body's immune defenses by regulating the function of immune cells. For example, baicalin and baicalin in *Scutellaria baicalensis* have been shown to regulate T cell activation and differentiation, which in turn affects the immune response [37].

In terms of antimicrobials, *Scutellaria baicalensis* extract may inhibit bacterial growth through a variety of mechanisms [38]. On the one hand, its components may act directly on the bacterial cell wall or cell membrane, disrupting the structural integrity of the bacteria, thereby inhibiting the growth and reproduction of bacteria. On the other hand, *Scutellaria baicalensis* extract may also enhance the body's ability to clear

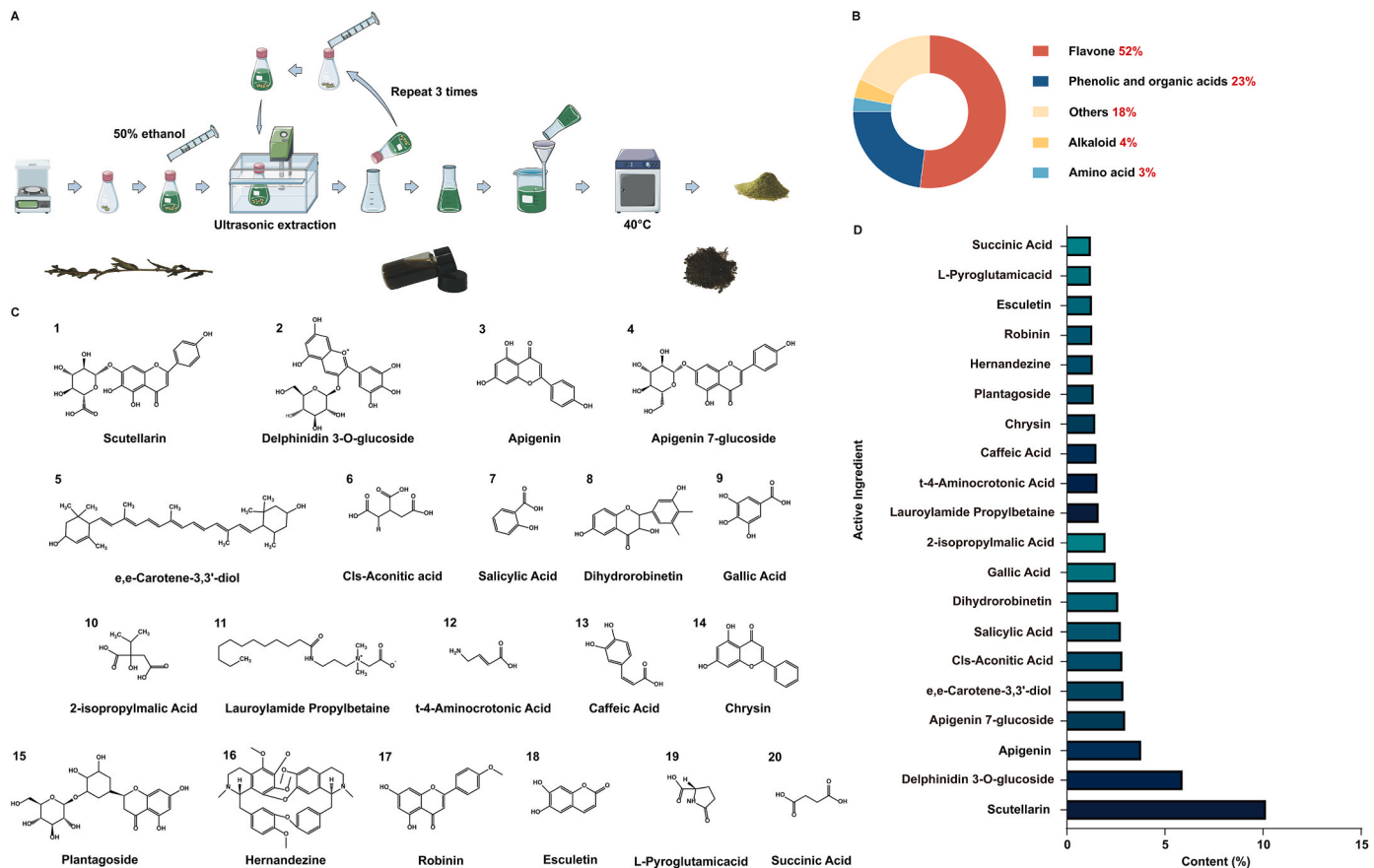


Fig. 6. Active ingredients from *Scutellaria baicalensis* stem and leaf. (A) Process flow diagram of *Scutellaria baicalensis* stem and leaf extraction. (B) The ratio of active ingredients in *Scutellaria baicalensis* stem and leaf. (C) Molecular formulas of the top 20 active ingredients in *Scutellaria baicalensis* stem and leaf. (D) Contents of the top 20 active ingredients in *Scutellaria baicalensis* stem and leaf.

Table 2
Scutellaria baicalensis stem and leaf phenolic acids and organic acid compounds.

Name	Molecular formula	Molecular weight	Peak time	Peak area/min	Additive ion	Relative content
Malic acid	C ₄ H ₆ O ₅	133.0140	1.2624	11042171	[M - H] ⁻	4.974 %
Cis-aconitate	C ₆ H ₆ O ₆	173.0092	4.5322	6288234.5	[M - H] ⁻	2.833 %
Salicylic acid	C ₇ H ₆ O ₃	137.0244	5.716116	6117527	[M - H] ⁻	2.756 %
Gallic acid	C ₇ H ₆ O ₅	169.0143	2.4625	5522575.5	[M - H] ⁻	2.488 %
2-Isopropylmalic acid	C ₇ H ₁₂ O ₅	175.0612	5.811467	4384056.5	[M - H] ⁻	1.975 %
3,4-Dihydroxycinnamic acid	C ₉ H ₈ O ₄	179.0350	6.87525	3359394	[M - H] ⁻	1.513 %
Succinic acid	C ₄ H ₆ O ₄	117.0193	1.763617	2691818.5	[M - H] ⁻	1.213 %
Veratric Acid	C ₉ H ₁₀ O ₄	181.0506	4.063334	1753582.625	[M - H] ⁻	0.790 %
Citric acid	C ₆ H ₈ O ₇	191.0190	4.457933	1424576.75	[M - H] ⁻	0.642 %
Vanilla acid O-sulfonate	C ₁₄ H ₁₈ O ₁₂ S	409.0442	5.614017	1410386.125	[M - H] ⁻	0.635 %
Rheic Acid	C ₁₅ H ₈ O ₆	283.0248	11.76115	1141842.75	[M - H] ⁻	0.514 %
Shikimic acid	C ₇ H ₁₀ O ₅	173.0450	1.110983	1059693.875	[M - H] ⁻	0.477 %
Vanillic acid	C ₈ H ₈ O ₄	167.0000	5.0158	1039517.063	[M - H] ⁻	0.468 %
Methylsuccinic Acid	C ₅ H ₈ O ₄	131.0345	3.624367	1010012.688	[M - H] ⁻	0.455 %
Malvidin	C ₁₇ H ₁₄ O ₇	331.0810	11.62868	620027.25	[M+H] ⁺	0.279 %
4-Hydroxybenzoic acid	C ₇ H ₆ O ₃	139.0390	1.2143	602061.75	[M+H] ⁺	0.271 %
Maltol	C ₆ H ₆ O ₃	127.0390	4.188933	601524.375	[M+H] ⁺	0.271 %
Isoeugenol	C ₁₀ H ₁₂ O ₂	163.0765	12.4192	421100.8438	[M - H] ⁻	0.190 %
4-Hydroxyphenylacetic acid	C ₈ H ₈ O ₃	151.0401	5.1278	389059.4375	[M - H] ⁻	0.175 %
Caffeic acid	C ₉ H ₈ O ₄	181.0495	5.364533	245053.625	[M+H] ⁺	0.110 %
Sinapic Acid	C ₁₁ H ₁₂ O ₅	225.0755	5.6917	204139.7344	[M+H] ⁺	0.092 %
Tretinoin	C ₂₀ H ₂₈ O ₂	299.2017	22.03338	170809.0938	[M - H] ⁻	0.077 %

bacteria by modulating the host's immune response. For example, *Scutellaria baicalensis* extract is able to enhance the phagocytic function of macrophages, promoting their phagocytosis and digestion of bacteria. In addition, *Scutellaria baicalensis* extract may also exert its pharmacological effects by regulating cell signaling pathways. For example, baicalein has been shown to activate the Nrf2 signaling pathway,

thereby enhancing the antioxidant capacity of cells and mitigating oxidative stress damage. This antioxidant effect may help protect host cells from oxidative damage caused by bacterial infection, thereby enhancing the body's antimicrobial capacity [39]. In summary, *Scutellaria baicalensis* extract may exert its antibacterial and immunomodulatory effects through a variety of molecular targets

Table 3
Scutellaria baicalensis stem and leaf alkaloid compounds.

Name	Molecular formula	Molecular weight	Peak time	Peak area/min	Additive ion	Relative content
Lauramidopropyl betaine	C ₁₉ H ₃₈ N ₂ O ₃	343.2955	16.83637	3601576	[M+H] ⁺	1.622 %
Hernandezine	C ₃₉ H ₄₄ N ₂ O ₇	653.3221	24.98083	2898291.5	[M+H] ⁺	1.306 %
Trigonelline	C ₇ H ₇ NO ₂	138.0550	1.0167	1950126.25	[M+H] ⁺	0.879 %
Pilocarpine	C ₁₁ H ₁₆ N ₂ O ₂	209.1285	2.557133	592162.375	[M+H] ⁺	0.267 %
Isorhynchophylline	C ₂₂ H ₂₈ N ₂ O ₄	383.1976	26.12367	245205.1406	[M – H] [–]	0.110 %

Table 4
Scutellaria baicalensis stem and leaf amin acid compounds.

Name	Molecular formula	Molecular weight	Peak time	Peak area/min	Additive ion	Relative content
L-Pyroglutamic acid	C ₅ H ₇ NO ₃	128.0353	1.555117	2709396.5	[M – H] [–]	1.221 %
Glutathione S-(4-nitrobenzyl)	C ₁₇ H ₂₂ N ₄ O ₈ S	443.1228	17.40713	1555137.875	[M+H] ⁺	0.701 %
Tyrosine	C ₉ H ₁₁ NO ₃	182.0812	1.3715	904974.9375	[M+H] ⁺	0.408 %
Acetyl Phenylalanine	C ₁₁ H ₁₃ NO ₃	206.0815	9.14245	532457.5625	[M – H] [–]	0.240 %
Methoxy group Tyrosine	C ₁₀ H ₁₃ NO ₄	210.0766	6.87525	233934.7813	[M – H] [–]	0.105 %

Table 5
Scutellaria baicalensis stem and leaf other compounds.

Name	Molecular formula	Molecular weight	Peak time	Peak area/min	Additive ion	Relative content
Tunaxanthin	C ₄₀ H ₅₆ O ₂	568.4275	26.3702	6409497.5	[M] ⁺	2.887 %
4-Aminocrotonic Acid	C ₄ H ₇ NO ₂	102.0550	1.3715	3465902.25	[M+H] ⁺	1.561 %
Erucylamide	C ₂₂ H ₄₃ NO	338.3415	21.88638	2487259	[M+H] ⁺	1.120 %
Glycerol palmitate	C ₁₉ H ₃₈ O ₄	331.2840	20.78732	2228330.25	[M+H] ⁺	1.004 %
6-Gingerol	C ₁₇ H ₂₆ O ₄	295.1901	17.31432	2135230.25	[M+H] ⁺	0.962 %
Myristic acid	C ₁₄ H ₂₈ O ₂	227.2017	21.43575	1711165	[M – H] [–]	0.771 %
Ricinoleic acid	C ₁₈ H ₃₄ O ₃	297.2435	21.31577	1116605.75	[M – H] [–]	0.503 %
Glycidaldehyde	C ₃ H ₆ O ₃	89.0244	0.9939833	1109902.125	[M – H] [–]	0.500 %
D- (–)-Arabinose	C ₅ H ₁₀ O ₅	149.0455	1.05285	1007908	[M – H] [–]	0.454 %
Sphingosine	H ₃ O ₄ P	96.9696	1.2624	1005261.625	[M – H] [–]	0.453 %
Metformin	C ₉ H ₁₁ NO ₃	182.0812	1.3715	904974.9375	[M+H] ⁺	0.387 %
Stearic acid	C ₁₈ H ₃₆ O ₂	283.2643	26.09515	619794.875	[M – H] [–]	0.279 %
4-Guanidinobutyric acid	C ₅ H ₁₁ N ₃ O ₂	146.0924	1.345383	606390.375	[M+H] ⁺	0.273 %
Coumarin	C ₉ H ₆ O ₂	147.0438	13.49172	590488	[M+H] ⁺	0.266 %
Indole-3-carboxylaldehyde	C ₉ H ₇ NO	146.0606	9.615717	557808.75	[M+H] ⁺	0.251 %
N-oleoyl ethanolamine	C ₂₀ H ₃₉ NO ₂	326.3051	22.6272	520426.0313	[M+H] ⁺	0.234 %
Vitamin B2	C ₁₇ H ₂₀ N ₄ O ₆	377.1456	6.63555	508950.75	[M+H] ⁺	0.229 %
Glyceride	C ₃ H ₆ O ₄	105.0193	1.05285	493920.8438	[M – H] [–]	0.223 %
Palmitamide	C ₁₆ H ₃₃ NO	256.2635	17.6967	489118.25	[M+H] ⁺	0.220 %
Sucrose	C ₁₂ H ₂₂ O ₁₁	341.1089	0.9939833	244506.125	[M – H] [–]	0.110 %
p-Coumaryl alcohol	C ₉ H ₁₀ O ₂	149.0608	1.081583	23999.4082	[M – H] [–]	0.011 %

and cell signaling pathways. The in-depth study of these mechanisms not only helps to elucidate the pharmacological effects of *Scutellaria baicalensis* extract, is conducive to the resource development of *Scutellaria baicalensis* stems and leaves, but also provides a theoretical basis for the development of new antibacterial drugs and immunomodulators.

CRedit authorship contribution statement

Xilin Wang: Visualization, Methodology, Conceptualization. **Shihai Wu:** Writing – original draft, Methodology, Conceptualization. **Ning Guo:** Visualization, Formal analysis. **Feike Yu:** Writing – review & editing, Data curation. **Xiaofeng Xu:** Writing – original draft, Methodology. **Xinghai Wang:** Data curation, Formal analysis. **Xiaohan Yu:** Writing – review & editing, Project administration. **Xiaoye Liu:** Visualization, Project administration. **Hong Dong:** Project administration, Funding acquisition, Conceptualization.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request. The dataset supporting the conclusions of this article are included within the article.

Ethics statement

The animal study was reviewed and approved by Genentech Institutional Animal Care and Use Committee at the Beijing University of Agriculture with its ethics approval number BUA2023117 and followed the ARRIVE guide. The experiment site was provided by Beijing Shuangyin Farm and all methods were performed in accordance with the relevant guidelines and regulations.

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Declaration of competing interest

The authors declare no competing interests.

Abbreviations

HQ	<i>Scutellaria baicalensis</i> (Huang Qin)
HQSL	<i>Scutellaria baicalensis</i> stem and leaf (Huang Qin stem and leaf)
UHPLC-MS/MS	ultra-high performance liquid chromatography tandem mass spectrometry
ANOVA	Analysis of Variance
PCR	Polymerase Chain Reaction
NCBI	National Center of Biotechnology Information
MEGA-X	Molecular Evolutionary Genetics Analysis

Data availability

The data that has been used is confidential.

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